CE Cyber Data Acquisition / Synthetic Data Generation Report

Purpose:
This report details how the research data was obtained or generated, its characteristics, quality controls, and any preprocessing or transformations performed prior to experimentation. It ensures the data used in analysis is transparent, ethical, and replicable.

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1. Project Context

Field	Response
Student Name(s)	Teaundra Thompson
Research Title	IDS Resiliency Against Evasion Attacks
Faculty Advisor	Professor Henderson
DoD/ACE Mentor	Professor Littleton
Version	v1.5
Date	12/15/2025

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2. Purpose of Data in Research

Describe how the data supports the research question and hypothesis.

Include:

- What is the role of the data (e.g., training, testing, validation, simulation)?
- How does the data relate to the independent/dependent variables in your study?
- What is the expected outcome from using this dataset?

The data that I will be using is the UWF-ZeekData24 dataset and pcap files that I create. The UWF-ZeekData24 dataset will be used to train an MLP ML model to identify attacks. The entire UWF-ZeekData24 dataset consists of 8 CSV files named: Benign, Credential_Access, Defense_Evasion, Exfiltration, Initial_Access, Persistence, Privelege_Escalation, and Reconnaissance. For the sake of research, I will only be using six of the previously said files, Persistence, Initial_Access, Benign, Reconnaissance, Credential_Access, and

Privilege_Escalation. The dataset relates to the independent variables because it establishes the baseline detection rate and overall accuracy of the model. The testing results utilizing 20% of the dataset is related to the dependent variables because the test results are dependent on the baseline accuracy of the MLP model. The MLP model is dependent on the IDS, Zeek, because it interprets Zeek's logs and generates analysis. My expected outcome from using this dataset will be a moderately accurate baseline IDS with an accuracy of no less than 70%. The pcap files will be an independent variable that affects the dependent variables. Depending on how I modify the pcap files they may increase or decrease the detection rate.

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3. Data Type and Source

Data Type	Description	Source / Provider	Access Method	Licensing / Ethics
Simulate	Dataset created in a controlled environment; 6 CSV files used	University of West Florida	Online repository, files were downloaded from the following link: https://datasets.uwf.edu/data/UWF-ZeekData24/csv/	Creative Commons Attribution 4.0 International
Synthetic	Zeek-generated log (conn.log)	Python script	Locally generated	Internal use
Simulated	Pcap files created in a vm envrionement	WireShark	Locally generated	Anonymized
Simulated	Pcap files malware analyzed in sandbox	Recorded Future Triage	Online	Publicly available

Note: Examples included in above table must be replaced with your data.

Include:

- Whether data is real-world, synthetic, simulated, or hybrid
- Origin of the data (organization, publication, or tool)
- Access method (API, file download, generator script)
- Data sharing/license terms
- IRB or ethical considerations (if applicable)

Tip: If synthetic data was generated, explain why real data was unavailable or why simulation was chosen.

Simulation was chosen to ensure there is no sensitive data present in pcap files. By generating them myself in a monitored network environment I can control what appeared in the pcap. Additionally, retrieving pcaps from Recorded Future Triage also ensures there are no PII due to the pcaps being generated in a sandbox. Synthetic data is generated due to the use of simulated data. The simulated pcap files are fed into the IDS, thus creating synthetic logs.

Note: The MLP model will simply produce output in the terminal of the Colab environment after analyzing the conn.log from Zeek to reduce file clutter. Screenshots will be taken of MLP output and placed in a respective folder for each test case.

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4. Data Acquisition / Generation Process

Describe how data was collected, captured, or generated.

Include:

- Hardware or software tools used
- Collection frequency (intervals, duration)
- Variables collected (features, labels, timestamps)
- Scripts or algorithms used for data generation

Step	Process Description	Tool / Script	Duration / Frequency	Output
1	Generate_pcaps/retrieve_pcaps from Recorded Future	Wireshark/Recorded Future	5 times	.pcap files
2	Feeding pcap into Zeek	Zeek	5 times	conn.log
3	MLP analysis of conn.log	MLP model	5 times	Terminal output

Note: For the pcap files that were generated I utilized wireshark within the vm environment. I engaged in light web browsing and I then used Scapy to modify the pcap to create evasion attacks within the pcap. After creating and modifying the pcaps I store it in file path "/Desktop/testCases/test_case". "test_case" will be the name of the current testcase. Then via terminal, I feed the specified pcap file into zeek, and the connection log (conn.log) file is generated in the same folder where the pcap resides. Then I take the conn.log file and upload it into my Colab notebook environment. I preprocess the log and add additional columns and then it is analyzed. The MLP produces output in the built-in terminal.

5. Data Characteristics

Provide a high-level summary of dataset structure and properties.

Attribute	Description
Data format	UWF-ZeekData24 dataset. Six CSV files concatenated into one dataset: Persistence, Initial_Access, Benign, Reconnaissance, Credential_Access, and Privilege_Escalation.
Number of records	95,522
Number of features	26
Labeled/Unlabeled	Labeled
Time range covered	n/a
Sampling frequency	One second frequency count
Data size	21.44 MB
Storage location	Local host machine & Zeek VM

Note: Examples included in above table must be replaced with your data.

Include:

- Schema or field names (example snippet)
- Labeling methods (manual, automated, semi-supervised)
- Examples of data entries (with sensitive data removed)

6. Pre-Processing Steps

Describe all transformations, cleaning, and feature engineering performed before data use.

Include:

- Normalization/scaling
- Data balancing (e.g., SMOTE)
- Feature selection/extraction
- Noise removal or outlier handling
- Data partitioning (train/test/validation)

Step	Operation	Tool / Library	Purpose	Output
1	Define_column names_and_identify features	Google Colab/Python	Make the dataset interpretable	Cohesive dataset in readable format for MLP
2	Scale values	Google Colab/Scikit-learn	Normalize input	Uniform range
3	Split train/test	Google Colab/ Scikit-learn	Validation	80/20 split
4	Preprocessing for conn.logs	Google Colab/ Scikit-learn	Make logs interpretable for MLP	Interpretable log for MLP

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7. Data Quality & Validation

Document how data accuracy and consistency were verified.

Include:

- Validation metrics (missing values, duplicates, anomalies)
- Ground-truth checks
- Random sampling review
- Bias analysis (if applicable)

Note: This table describes the preprocessing for the UWF-ZeekData24 dataset

Metric	Description	Result
Missing Values	% of null entries	0%
Duplicate Records	% duplicates	1%
Class Imbalance	Ratio of majority to minority class. Majority class is "normal" and minority class is "privilege escalation/persistence".	23,953:163

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8. Ethical and Security Considerations

Include:

- Data anonymization steps (e.g., hashing, tokenization)
- Compliance with DoD, FERPA, or IRB standards
- Confirmation of synthetic or anonymized data (no PII/CUI)
- Secure storage and access control measures

Zeek's live network monitoring feature will be disabled to prevent capture of PII or CUI. Pcap files will be generated within the VM environment and sourced from Recorded Future Triage. The pcaps will be manually checked to ensure there are no unencrypted protocols like HTTP present. The UWF-ZeekData24 dataset is publicly available from the University of West Florida.

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9. Data Sampling & Collection Intervals

Describe when and how often data was captured or simulated.

Parameter	Description
Pcap collection	Will be done 7 times for each test case (7 test cases)
Duration of collection	3 minutes or less for each capture. Ensures a decent size pcap capture to make evasion attacks discrete, and allows for lower time duration for high volume attacks such as

	DDoS (such a high volume attack could have adverse effects on the testbed if executed for an extended duration).
Time window(s)	Varies: dependent on when I simulate traffic and capture it. No specific time window required.

Tip: If time-series data, include temporal coverage and granularity.

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10. Data Storage & Management

Include:

- Storage location and access path
- File naming conventions
- Backup procedures
- Repository links (Git, Jira, or shared drive)
- Data versioning strategy

All test case files will be stored in their respective files paths on the vm "/Desktop/testcases/TestCaseID". Subsequently, the "testCase" folder will be uploaded to GitHub in repository ACE-Senior-Project upon completion of all test cases. Pcap files will be named according to the type of traffic or attacks they simulate. After a test case is completed, a snapshot will be taken to ensure the VM state does not become corrupted. When a new snapshot is taken, the previous one will be deleted to prevent excessive use of hardware resources.

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11. Limitations and Known Issues

Include:

- Incomplete features or missing labels
- Limited representativeness (e.g., small dataset size)
- Known noise or bias sources
- Synthetic realism limitations

Example:

Synthetic user activity patterns may not fully replicate adversarial behavior seen in production networks.

The UFW-ZeekData24 dataset is heavily imbalanced which may affect anomaly classification even after calculating class weights to handle imbalance. Also, due to the testbed being in an isolated environment the experiment will not fully replicate adversarial evasion in a typical network with copious traffic.

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12. Summary Table

Section	Key Content
Data Type	UFW-ZeekData24 dataset
Source	https://datasets.uwf.edu/data/UWF-ZeekData24/csv/
Volume	95,522 samples
Pre-processing	Identify features, calculate class weights for balancing
Validation	Test accuracy > 90%
Security	Anonymized
Storage	Host machine and Zeek VM
Use	Model training and testing

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13. Mentor Review / Approval

Reviewer	Role	Date	Notes
Student Peer			
Faculty Advisor			
DoD/ACE Mentor			

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Submission Guidance

Discuss with assigned ACE Team Member prior to submission of any datasets to determine expected process and necessary artifacts (e.g. scripts, notebooks, documents, etc.)

- Attach datasets or sample excerpts (if size allows).
- Include any scripts or notebooks used for generation or cleaning.
- Provide references or DOIs for any public datasets.
- Submit as .docx or .pdf, with code or data links as appendices.

Summary — Required Sections

There is no action required for this section. This summary section is merely to provide understanding of each section's importance toward your R&D project completion.

	Section	Why It Matters
1	Project Context	Traceability
2	Purpose	Alignment with research goals
3	Source	Provenance and ethics
4	Process	Reproducibility
5	Characteristics	Transparency
6	Pre-Processing	Data readiness
7	Quality	Credibility
8	Security	Compliance
9	Sampling	Statistical rigor

10	Storage	Data management
11	Limitations	Honesty and scope
12	Summary	Transparency
13	Review	Oversight

Instructions

This instructional guide is designed to assist in the completion of the template as a means to help students understand expectations, rationale, and best practices for each section within this document.

Overview:

This deliverable provides a structured, standardized way for students to document how they obtained, generated, prepared, and validated their research data. It ensures that research data practices are transparent, ethical, reproducible, and technically defensible, aligning with DoD and ACE Cyber R&D requirements.

Students should treat this as both a technical record (for reproduction) and an ethical declaration (for compliance).

1. Project Context: Self-explanatory

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2. Purpose of Data in Research:

Students must describe why data is required and how it supports their research question or hypothesis.

Instructions:

- Clarify what role the dataset serves: is it used for training, testing, validation, or simulation?
- Relate the dataset directly to the independent and dependent variables in the research design.
- Explain what insights or outcomes are expected from analyzing this data.
- Provide clear linkage between data purpose and the hypothesis.

Example: “The dataset is used to test the effectiveness of a false-positive reduction algorithm in an Elastic SIEM by simulating real alert patterns.”

Why is this important? It ensures that data usage aligns directly to the research purpose and avoids “data for data’s sake.”.

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3. Data Type and Source:

Students identify where their data originates and classify it properly.

Instructions:

- Specify whether the data is real-world, synthetic, simulated, or hybrid.
- Clearly document the origin — include organization, repository, or script name.
- Describe how it was accessed (download, API, lab tool).
- Declare any data licensing or ethics requirements, including IRB considerations.

Helpful Tips:

- ☐ If using synthetic data, justify why real data wasn’t used (e.g., security restrictions, PII concerns).
- ☐ Use the table format provided; replace all example rows with actual project entries.

Why is this important? It demonstrates provenance, ethical compliance, and reproducibility.

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4. Data Acquisition / Generation Process:

This section documents exactly how data was captured or generated, making it possible to reproduce.

Instructions:

- Provide step-by-step descriptions of how each dataset or log file was collected or created.
- List all tools, scripts, or systems used (e.g., Python scripts, Wireshark, Kibana).
- Include duration and frequency of collection (e.g., every 10 min for 24 hours).
- Describe outputs produced at each step.

Best Practices:

- ☐ If using automated generators, include references to code, function names, or GitHub script paths.
- ☐ Why is this important? It ensures methodological transparency and allows future replication.

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5. Data Characteristics:

Students provide a snapshot of the data structure, scope, and format.

Instructions:

- Fill out each attribute (e.g., data format, record count, size).
- Include whether the dataset is labeled or unlabeled, and if labeled, how labeling was done.
- Give examples of schema fields or column names (non-sensitive).
- Specify where and how the data is stored (local VM, Git repo, S3, etc.).

Why is this important? Provides reviewers with a concise view of the dataset's scope, complexity, and readiness.

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6. Pre-Processing Steps:

Students describe all actions taken to clean, transform, or prepare the data for modeling.

Instructions:

- Use the provided table to document each transformation (e.g., removing nulls, scaling).
- List tools or libraries used (e.g., Pandas, NumPy, Scikit-learn).
- Describe why each step was needed and what output was produced.
- If applicable, include code snippets or Jupyter notebook references as appendices.

Why is this important? Demonstrates data hygiene, scientific rigor, and traceability of preprocessing.

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7. Data Quality & Validation:

Students validate their dataset for accuracy, completeness, and balance.

Instructions:

- Calculate key quality metrics such as:
  - o % of missing values o %
  - duplicates o Class
  - imbalance ratios
- Document ground-truth or spot checks performed.
- Note any bias or inconsistencies detected.

- Include validation scripts or summaries in appendices if available.

Example: “Timestamps verified to fall within 30-day study window; 0.8% missing entries addressed via interpolation.”

Why is this important? It ensures dataset integrity and supports trust in the findings.

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8. Ethical and Security Considerations

Students confirm data compliance and confidentiality measures.

Instructions:

- Explain how PII/CUI was handled or anonymized (hashing, tokenization, etc.).
- Confirm compliance with DoD, FERPA, or IRB guidelines.
- Describe secure storage practices (encryption, access control).
- If synthetic, state that no real user data was included.

Why is this important? It protects the research team, University, and sponsors (DoD/NIWC) from privacy and compliance violations.

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9. Data Sampling & Collection Intervals:

Students explain the temporal structure of their data.

Instructions:

- Define sampling frequency (e.g., 1s, 5min, hourly).
- Note total collection duration and any specific time windows (e.g., working hours).
- Quantify number of samples or records gathered.

Why is this important? It ensures the dataset is representative and statistically valid.

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10. Data Storage & Management:

Students describe how data is stored, organized, and versioned.

Instructions:

- Identify storage location (e.g., ACE Lab Drive, GitLab).
- Explain naming conventions and directory structure.
- Describe backup or redundancy procedures.
- Include versioning details (e.g., commit hashes, version folders).

Why is this important? Promotes data integrity, traceability, and collaboration.

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11. Limitations and Known Issues:

Students demonstrate critical awareness of their dataset's imperfections.

Instructions:

- Identify incomplete or noisy data.
- Note if sample size or diversity is limited.
- Describe biases introduced by synthetic generation.
- Acknowledge simulation limitations.

Example: "Generated login failures may not fully reflect adversary tradecraft patterns."

Why is this important? Adds academic honesty and guides future improvement.

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12. Summary Table:

Students create a one-page summary of all key data details.

Instructions:

- Fill out each field concisely (e.g., source, type, size, validation results).
- This serves as a quick reference for reviewers and mentors.

Why is this important? Aids in rapid review and cross-comparison between projects.

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13. Mentor Review / Approval:

Students capture feedback and validation from mentors and advisors.

Instructions:

- Leave blank until reviewed.
- Faculty and ACE mentors should provide approval and comments before final submission.

Why is this important? It establishes accountability and quality control.

Submission Guidance:

Before you submit…Consult your ACE mentor regarding dataset submission expectations to discuss.

- ☐ Attaching sample data excerpts or include links to repositories.
- ☐ Including scripts or notebooks used for generation or preprocessing.
- ☐ Export submission of .docx or .pdf, with appendices for code/data.